Unit 202 — Four-Asset Pilot Review

Package: CC-11.9 · Generated: 2026-08-24T23:24:23.492Z

NOT ALL FOUR PASS — NO-GO, STOPPED BEFORE BULK GENERATION

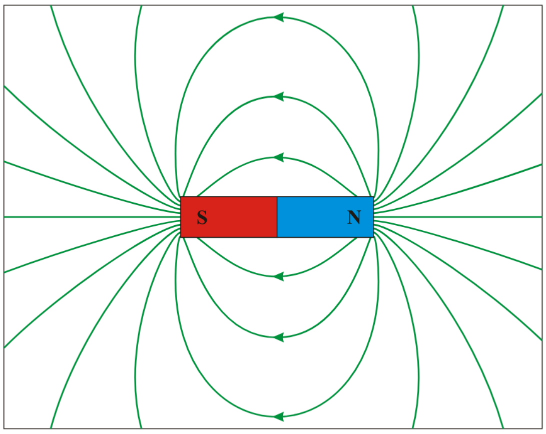
Assets attempted	4
PASS	2
RETRY (mid-pipeline, superseded by a later attempt)	0
HUMAN_REVIEW_REQUIRED	1
Not completed	1

Bar magnet and magnetic field

unit202.magnet.field

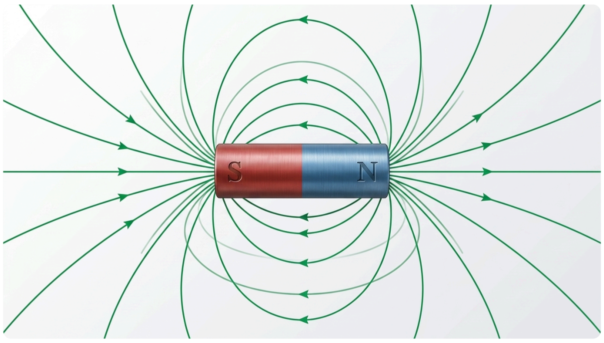
Curriculum context	LO5 — lesson.magnetism.fundamentals
Pedagogical role	PHENOMENON
Production class	HYBRID
Purpose	Show a bar magnet with its external magnetic field pattern, N to S, as the basis for flux/flux-density teaching.

REFERENCE (AS SENT TO GEMINI)



Wikimedia Commons — DipolMagnet.svg

FINAL ALP ARTWORK (ATTEMPT 3)



Model: gemini-3.1-flash-image · Generated: 2026-08-24T23:18:56.589Z

AUDIT VERDICT: HUMAN_REVIEW_REQUIRED

Immutable fact	Result	Note
meaningful field-line geometry	PASS	Field-line loops are present again, correctly nested above and below the bar, matching the reference's tiered arc structure.
external field direction runs N to S	UNCERTAIN	The nested ARC loops are directionally correct: every arc's peak arrowhead points toward S (left, matching this image's own S-on-left/N-on-right orientation, same as the reference), consistent with field exiting N, arcing over, and entering S. HOWEVER: direct side-by-side comparison with the cached reference (unit202.magnet.field.raster.png) shows the reference deliberately draws its two straight horizontal axis rays (extending left from S and right from N to the frame edge) WITH NO ARROWHEADS AT ALL. This generation invented arrowheads on both horizontal rays that the reference does not have. The right (N-side) ray's invented arrow points outward/rightward, which happens to be directionally consistent with exiting N -- but the left (S-side) ray's invented arrow also points outward/leftward (away from the magnet), which is backward for a field that should be entering S from outside. This is exactly the defect class the corrected asset contract explicitly prohibits ("do not invent arrowheads on cropped/incomplete field lines" / "no arrowheads added to partial lines unless their correct direction is unambiguous and explicitly permitted").
field-line density used meaningfully where density is taught	PASS	Not applicable to this state (density_comparison=false).

Prohibited change	Result	Note
do not draw field lines that reverse direction or cross incorrectly	FAIL	The invented arrowhead on the S-side horizontal ray points away from the magnet (outward), the reverse of the direction a field line entering S should point. This is a genuine, confirmable directional error, not a matter of interpretation -- verified by direct comparison against the reference's own (unarrowed) horizontal rays.

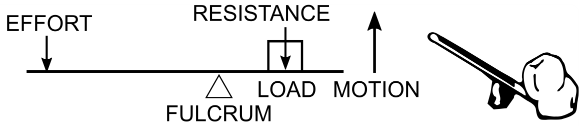
Overall findings	Retry budget exhausted (1 initial + 1 correction, both used). The image is substantially improved and close to correct -- N/S labels present and correctly placed, field-line topology and arc-loop directions correct, white background correct -- but a specific, real, confirmable defect remains: an invented arrowhead on the S-side horizontal ray points the wrong way (outward instead of into S), a defect the reference image itself avoids by simply not arrowing that line at all. Per the task brief's bounded-retry rule, this is not run a third time. Marked HUMAN_REVIEW_REQUIRED rather than PASS or FAIL: the core geometry is largely correct and the residual defect is narrow and specific, but 'looks close' is not the same as verified-correct, and an incorrect directional arrow on an assessment-adjacent asset is not something to auto-promote. A human (Product Owner/ChatGPT) should judge whether this is acceptable as-is, needs the arrow manually removed/corrected, or needs a further prompt-engineering pass explicitly forbidding any arrowhead on the two straight axis rays.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.magnet.field\unit202.magnet.field-master-v3.png`

Lever — Class I

unit202.levers.class-1

Curriculum context	LO3 — lesson.foundation.physics.simple-machines
Pedagogical role	CONFIGURATION
Production class	HYBRID
Purpose	Show a Class I lever arrangement (fulcrum between effort and load) so a learner can recognise it from the arrangement itself.

REFERENCE (AS SENT TO GEMINI)



Pearson Scott Foresman — Lever (PSF).svg

FINAL ALP ARTWORK (ATTEMPT 3)

Model: gemini-3.1-flash-image · Generated: 2026-08-24T23:20:43.588Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
fulcrum positioned between the effort point and the load point	PASS	Single lever bar with the fulcrum wedge clearly positioned between the EFFORT end (left, with a hand-grip loop) and the LOAD end (right, with a cube load). Correct Class I ordering.

Prohibited change	Result	Note
do not blend this with the Class II or Class III arrangement	PASS	Exactly one lever diagram is present. No trace of the Class II or Class III sibling arrangements -- the prepared/cropped reference (isolating only the top subfigure) resolved the composite-reference defect that caused attempts 1-2 (CC-11.8) to fail on this exact point.

Overall findings	Strong PASS on first attempt of this production run. The composite-reference crop (isolating the Class I subfigure before sending to Gemini) fully resolved the defect that blocked this asset in the earlier CC-11.8 proof (which had sent the full 3-class composite and got a blended result). No immutable fact violated, no prohibited change made, required labels present and correct.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.levers.class-1\unit202.levers.class-1-master-v3.png`

Simple rotating-loop AC generator — loop plane facing poles — near-zero EMF

unit202.generator.rotating-loop.horizontal

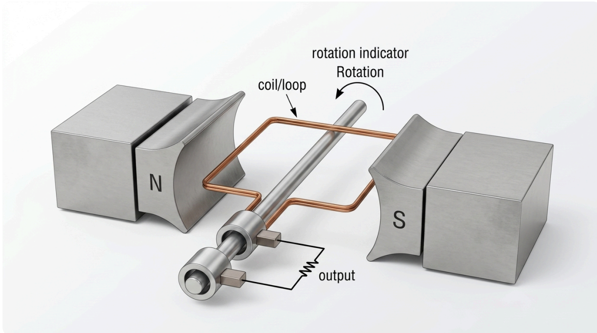
Curriculum context	LO5 — lesson.emf.ac-generation-principles
Pedagogical role	PHENOMENON
Production class	HYBRID
Purpose	Show a single loop of wire, loop plane facing the poles head-on (widest visible loop face), rotating on a central axis between N and S poles, establishing the physical basis of single-loop AC generation at Level 2 depth.

REFERENCE (AS SENT TO GEMINI)

reference preview not found

Wikimedia Commons — Diagram of single-phase generator with two poles.svg

FINAL ALP ARTWORK (ATTEMPT 1)



Model: gemini-3.1-flash-image · Generated: 2026-08-24T23:21:37.579Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
N/S magnetic poles	PASS	N (left) and S (right) clearly labelled on the pole pieces.
loop/coil between the poles	PASS	Single copper loop visible, positioned between the N and S pole faces.
loop plane facing poles head-on (widest visible loop face) -- defining fact of this asset	PASS	The loop's rectangular face is substantially visible from this viewing angle (not an edge-on sliver), consistent with the near-zero-EMF, face-on pose requested.
central rotational axis	PASS	A single common shaft runs through the loop and both slip rings.
output/slip-ring concept at governed Level-2 abstraction	PASS	TWO separate slip rings are visible, mounted concentrically on the shaft, each contacted by its own stationary brush, with wires running from the brushes to a resistor labelled OUTPUT. This is a materially complete topology, not merely a vague 'concept' -- directly resolves the CC-11.9 handover's complaint about the prior catalogue's under-specified slip-ring wording.
rotating conductor cuts magnetic flux	PASS	Implied correctly by the loop-between-poles + rotation-indicator composition; not contradicted anywhere.

Prohibited change	Result	Note
do not substitute a detailed modern alternator	PASS	Kept at the elementary single-loop/two-slip-ring level matching the DOE reference, not an over-detailed modern machine.

Prohibited change	Result	Note
do not add three-phase windings, phasors or brushes/commutator detail beyond governed scope	PASS	Exactly two brushes, two rings -- no commutator (a commutator would be a single split ring; this is unambiguously two separate full rings), no three-phase winding.
do not depict the edge-on loop pose in this asset -- that is the sibling ProductionAsset	PASS	The loop is shown face-on, not edge-on -- correct pose for the horizontal/near-zero-EMF asset.

Overall findings	Strong PASS on first attempt. The prepared DOE Figure 1 crop (a genuine, complete, technically authoritative topology reference -- selected via real text search of the handbook for 'slip ring' rather than guessed) gave Gemini an unambiguous topology to redraw, and it preserved every required structural element: two separate slip rings, two brushes, correct output path, no commutator, correct N/S, correct face-on pose. This directly satisfies the CC-11.9 handover's specific concern that the previous generic Commons SVG reference under-specified the slip-ring mechanism.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.generator.rotating-loop.horizontal\unit202.generator.rotating-loop.horizontal-master-v1.png`

unit202.components.physical.resistor

No completed generation + audit found for this asset -- pending (e.g. blocked on an external reference-source rate limit at report time).